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Systematic Review

Artificial Intelligence Approaches to Energy Management in HVAC Systems: A Systematic Review

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Abstract: Heating, Ventilation, and Air Conditioning (HVAC) systems contribute a considerable share of total global energy consumption and carbon dioxide emissions, putting them at the heart of the issues of decarbonization and removing barriers to achieving net-zero emissions and sustainable development goals. Nevertheless, the effective implementation of artificial intelligence (AI)-based methods to optimize energy efficiency while ensuring occupant comfort in multifarious settings remains to be fully realized. This paper provides a systematic review of state-of-the-art practices (2018 and later) using AI algorithms like machine learning (ML), deep learning (DL), and other computation-based techniques that have been deployed to boost HVAC system performance. The review highlights that AI-driven control strategies can reduce energy consumption by up to 40% by dynamically adapting to environmental conditions and occupancy levels. Compared to other work that focuses on single aspects of HVAC management, this work deals with the methods of control and maintenance in a comprehensive manner. Rather than focusing on abstract applications of machine learning models, this study underlines their applicability in HVAC systems, bridging the science–practice gap. This study highlights the prospective role AI could play, on the one hand, by enhancing HVAC systems' incorporation, energy consumption, and building technologies, while, on the other hand, also addressing the potential uses AI can have in practical applications in the future, bridging gaps and addressing challenges.

Keywords: artificial intelligence (AI); energy management; heating, ventilation, and air conditioning (HVAC) system

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1. Introduction

With city growth, a rise in the number of factories and people, and improved living standards, energy use in buildings has sharply increased. Currently, buildings account for more than 40% of the world's annual energy consumption and are responsible for 40% of total carbon dioxide emissions [1,2]. Due to global climate change, we may experience more extreme weather in the future. Therefore, when designing buildings, it is crucial to consider both climate change and future energy demands [3]. Nearly 40% of total energy is consumed by HVAC systems, including air conditioners, heat pumps, ventilation, and refrigeration equipment, making them significant contributors to energy use in commercial buildings [4,5]. Consequently, reducing energy costs and the carbon footprint requires a

more efficient use of energy in buildings. Managing a building's energy consumption is complex due to the influence of various factors, such as the building's location, weather conditions, desired indoor temperature and humidity, the number of people present and their time spent indoors, the building's thermal insulation, and its purpose. Since most people spend the majority of their time indoors [6], ensuring indoor comfort is essential for quality of life and productivity.

For HVAC systems to effectively transfer heat and mass, they have to contain specific elements such as air supply ducts, heating/cooling coils, boilers, and chillers. Their operations will depend on conditions such as air temperature, flow velocity, and humidity. These features, among others, are said to be influenced by other external factors that include weather conditions, sunlight, and outdoor temperature. The American Society of Heating, Refrigerating and Air-Conditioning Engineers (ASHRAE) has extensively studied HVAC systems and their applications [6]. HVAC systems consume nearly half of a building's energy, so they must operate effectively to create a comfortable indoor environment. Enhancing the energy efficiency of HVAC systems has, therefore, become vital to improving the overall energy performance of buildings. Many researchers have worked on reducing the energy consumption of HVAC systems while maintaining occupant comfort.

The integration of Building Energy Management Systems (BEMSs) enables autonomous control of building activities, including HVAC [7], lighting, and equipment [8]. Studies suggest that, by employing intelligent technologies and analytics, office buildings can achieve 18% energy savings, while retail buildings can reach 12% [9]. BEMSs are designed to ensure the efficient operation of building systems, equipment, and services, thereby providing a more comfortable environment for occupants while reducing energy use and operational costs. BEMSs have greatly enhanced automated monitoring and control of HVAC systems, reducing the need for manual interventions and improving building efficiency, making them a key development in smart building technologies.

Different computer and software technologies bring more and more industries to automation; AI is playing a great role in it nowadays [1]. With the growing demand for optimization of energy usage, waste reduction, and maintaining comfort–security–health standards, the demand for smart buildings has appeared [2]. With the growing interest in building monitoring and control technologies like AI and the Internet of Things (IoT), the market for "smart buildings" is expected to expand rapidly. Both academia and industry are adopting AI in the building sector to improve design, construction, operations, and maintenance processes [8,9]. These could be AI algorithms merged with sensors that track and control indoor environmental conditions in real time. Such systems automatically analyze data, support predictions about performance into the future, and provide information to make informed decisions satisfactorily [3]. The McKinsey Global Institute [10] conducted a comparative study of AI adoptions across industries, further emphasizing the need for the building and construction sector to accelerate digital transformation and the adoption of AI. Investigating AI techniques is crucial for improving building energy efficiency and addressing related challenges, including the slow pace of AI adoption in this domain. There is a need for comprehensive analysis and evaluation of AI's current applications and its potential in future building solutions. Given rising global energy demands, HVAC systems have become focal points in urban energy management strategies. This paper examines recent advancements in AI technologies designed to enhance HVAC efficiency in line with stricter global standards and policy reforms. Advanced control methods with AI and HVAC optimization through maintenance management have been studied in recent times, and machine learning algorithms were found to be highly effective in optimizing energy efficiency [11,12]. Further, it has been demonstrated through studies that integrating smart

control algorithms with real-time analytics is very effective in optimizing HVAC, reducing energy consumption while keeping the interior comfortably cooled [13].

In this paper, we present a clear research hypothesis, demonstrate the novelty of our approach, and identify how our findings contribute to the field of AI-enhanced HVAC systems. Our hypothesis suggests that integrating advanced AI techniques, such as machine learning (ML) and deep learning (DL), into HVAC systems can improve energy savings while maintaining occupant comfort. This hypothesis shapes our research objectives and drives our comprehensive investigation into the application of these technologies across various building types.

Unlike previous studies that often focus on a single aspect of HVAC energy management, this research provides an integrated review of recent advancements in AI for HVAC systems. Our approach examines AI applications across multiple HVAC components, focusing on the dual goals of enhancing energy efficiency and operational effectiveness.

This paper primarily contributes to the academic and practical understanding of AI's potential to transform energy management within buildings. Unlike prior studies that focus on isolated aspects of HVAC management, this work provides a comprehensive overview of both control and maintenance techniques. Rather than focusing on the theoretical aspects of machine learning models, this study emphasizes their practical applications in HVAC systems, bridging the gap between research and implementation. As such, this work is a valuable resource for researchers and practitioners looking to implement AI solutions in pursuit of broader sustainability and energy efficiency goals. This study is a general overview of AI in HVAC optimization, with more emphasis laid on summing up methodologies, trends, and challenges than in listing specific case studies. Practical implementations are mentioned, however, so that readers can refer to real-world applications when needed.

This research offers a comprehensive and systematic literature review (SLR) on the integration of AI techniques in HVAC energy management. The novelty of this study lies in its rigorous synthesis of current research across diverse databases, highlighting AI-driven approaches for optimizing HVAC systems' energy efficiency while ensuring occupant comfort. Beyond merely reviewing existing literature, this study categorizes and assesses various AI algorithms and their applications within this domain, providing deeper insights into both control and maintenance strategies. Additionally, this research innovatively identifies significant research gaps and outlines potential future research directions, emphasizing the role of AI in enhancing both the operational efficiency and predictive maintenance of HVAC systems. Our systematic approach helps map the landscape of AI applications in HVAC energy management, making it a valuable resource for researchers and practitioners working toward improved building energy efficiencies. The significance of this research is further discussed in Section 5.

2. Review Methodology

2.1. Research Questions

An SLR process begins with defining the research questions (RQs). In this paper, we identify the following RQs:

RQ1. *What methods are available that use AI techniques to improve the energy performance of HVAC systems?*

RQ2. *What AI algorithms are used most widely in each method of improving HVAC energy efficiency?*

RQ3. *What are the reported outcomes and effectiveness of AI techniques in improving the energy efficiency of HVAC systems?*

2.2. Literature Search

In computer science and engineering, systematic literature reviews are commonly conducted using string-based searches in various digital libraries. Utilizing multiple databases to encompass a wide range of evidence during the database searching is strongly advised. As seen in Figure 1, this research searches the literature for pertinent contributions linked to the RQs outlined in Section 2.1 utilizing a database search based on three of the most used digital libraries for performing SLRs. Multiple databases, namely Science Direct, Google Scholar, and Scopus are discussed for conducting a systematic literature review (SLR). Since Science Direct has the highest level of coverage for an SLR, allowing it to facilitate reproducible and efficient searches through queries, filters, and hands-on searching, it is considered the main search engine. Unlike Scopus and Web of Science (WoS), Science Direct provides full-text access to several journals and books. Considering that there exists, in parallel, literature coming from other publishers, Google Scholar was utilized as a supplementary search tool, thereby retrieving roughly the same number of citations as WoS and Scopus. Scopus still had wider coverage of engineering records, which made it more appropriate to carry out evidence synthesis on engineering applications. Hence, Scopus was chosen as an additional specialized search database in the present study. Additionally, a forward and backward citation search was performed to ensure that all relevant studies, including those indexed in other databases, such as IEEE Xplore and Web of Science, were included. This approach ensured a comprehensive review of the available literature.

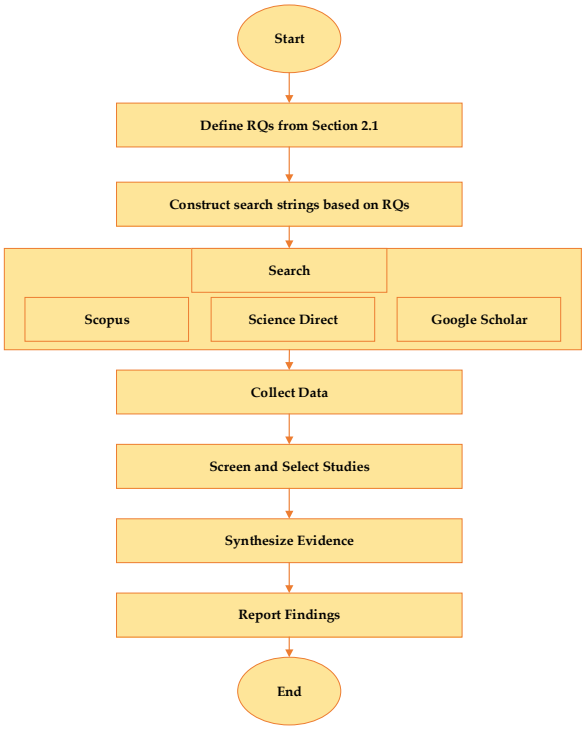


Figure 1. Search script flow diagram.

To conduct corresponding searches on the selected digital libraries described above, "Machine AND Learning" OR "Deep AND Learning" OR "AI" AND "Buildings AND Energy AND HVAC" search strings are defined based on the RQs introduced in Section 2.1.

2.3. Inclusion/Exclusion Criteria

The SLR undertaken in this study relies predominantly on peer-reviewed publications from reputable journals, books, and international conference proceedings, considered

reliable sources of scholarly literature. Additionally, relevant articles from the grey literature may also be incorporated. To emphasize the contemporary advancements and trends in HVAC energy management utilizing AI methods and optimization, only articles published within the six-year timeframe from 2018 and later are included throughout this paper.

2.4. Literature Search Results

A comprehensive search of academic literature was carried out utilizing Figure 2 to locate relevant articles from the three specified databases mentioned in Section 2.2, namely Science Direct, Scopus, and Google Scholar. The search yielded a total of 374 articles. Articles that concentrated on applying AI approaches to enhance HVAC energy management were included in the analysis. Only peer-reviewed journal articles, conference papers, and books published by respectable publishers were regarded as high-quality sources to guarantee the research's validity and legitimacy. The inclusion of such sources is illustrated in Table 1.

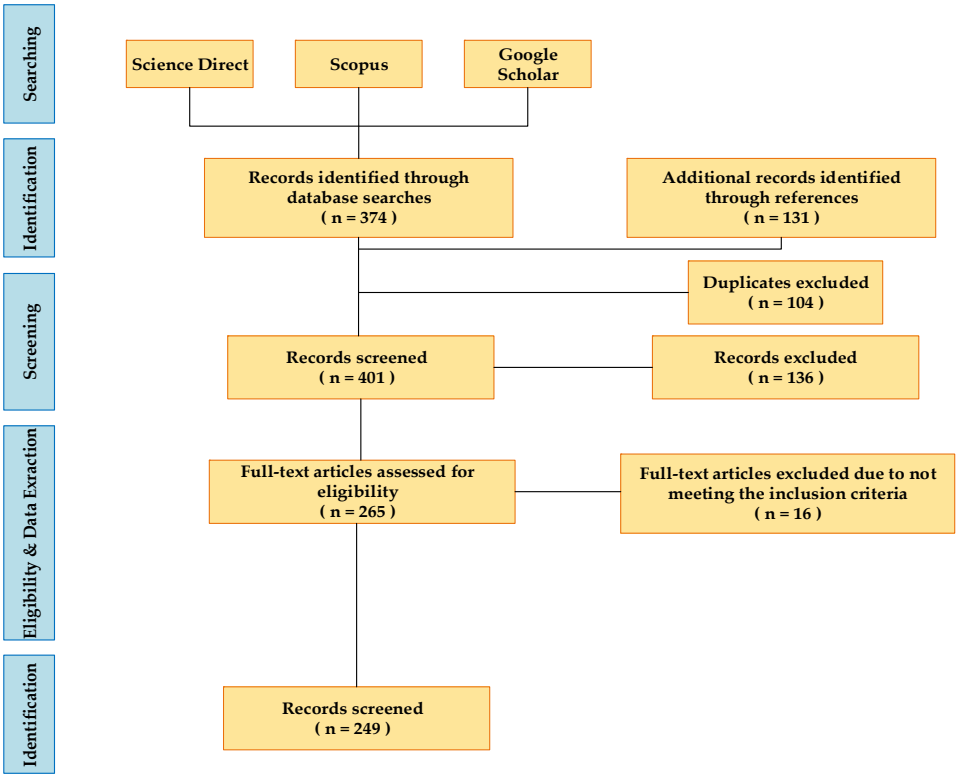


Figure 2. Literature search flow diagram.

Table 1. Top cited publications.

Title	Contribution	Importance	Novelty	Institution
Gradient boosting machine for modeling the energy consumption of commercial buildings [14]	Significant for modeling energy consumption	High impact on commercial building design	Introduces gradient boosting, an advanced machine learning technique, for energy modeling. This method offers a sophisticated approach compared to traditional models, improving prediction accuracy.	Lawrence Berkeley National Laboratory [USA]

Table 1. *Cont.*

Title	Contribution	Importance	Novelty	Institution
Using machine learning techniques for occupancy-prediction-based cooling control in office buildings [15]	Key for energy-efficient cooling control	Critical for reducing office energy usage	Applies machine learning techniques to predict occupancy and optimize cooling control, providing a modern solution for building management and improving energy efficiency.	ETH Zürich [Switzerland]
Whole building energy model for HVAC optimal control: A practical framework based on deep reinforcement learning [16]	Practical framework for HVAC control	High impact on HVAC optimization	Utilizes deep reinforcement learning for HVAC control, a cutting-edge method that offers a significant innovation in optimizing energy usage and system performance.	Carnegie Mellon University [USA]
Deep Reinforcement Learning for Smart Home Energy Management [17]	Advances smart home energy management	Important for residential energy efficiency	Applies deep reinforcement learning to optimize energy management in smart homes, introducing a novel approach to improving energy efficiency and control.	National Natural Science Foundation [China]
Predictive modeling for US commercial building energy use: A comparison of existing statistical and machine learning algorithms using CBECS microdata [18]	Comparison of various models	Significant for improving predictive accuracy	Compares statistical and machine learning models for energy prediction, providing insights into the effectiveness and advancements of newer algorithms in the field.	Northeastern University [USA]
Generative adversarial network for fault detection diagnosis of chillers [19]	Fault detection using advanced techniques	Essential for fault diagnosis in chillers	Innovatively applies generative adversarial networks (GANs) for diagnosing faults in chillers, a novel approach that enhances fault detection by leveraging GANs' unique capabilities.	National University of Singapore [Singapore]
Multi-Agent Deep Reinforcement Learning for HVAC Control in Commercial Buildings [20]	Multi-agent approach to HVAC control	Important for complex building systems	Introduces a multi-agent deep reinforcement learning approach for controlling HVAC systems, offering a sophisticated method for managing complex systems with multiple agents.	Nanjing University [China]

Table 1. *Cont.*

Title	Contribution	Importance	Novelty	Institution
Deep reinforcement learning to optimize indoor temperature control and heating energy consumption in buildings [21]	Optimization of temperature control	Crucial for energy savings in buildings	Uses deep reinforcement learning to optimize indoor temperature and heating energy consumption, presenting a novel and advanced method for improving building energy efficiency.	Politecnico di Torino [Italy]
Intelligent multi-zone residential HVAC control strategy based on deep reinforcement learning [22]	Multi-zone HVAC control strategy	High relevance for residential HVAC systems	Applies deep reinforcement learning to control multiple zones in residential HVAC systems, offering a new approach for enhanced and efficient multi-zone control.	University of Tennessee [USA]
A review of studies applying machine learning models to predict occupancy and window-opening behaviors in smart buildings [23]	Comprehensive review of machine learning	Important for understanding occupancy models	Provides a comprehensive review of machine learning applications for predicting occupancy and window-opening behaviors, summarizing significant advancements and methodologies.	Tianjin University [China]
Gnu-RL: A precocial reinforcement learning solution for building HVAC control using a differentiable MPC policy [24]	New reinforcement learning solution	Significant for optimizing HVAC control	Presents a novel reinforcement learning solution with a differentiable MPC policy for HVAC control, offering an innovative approach to managing building systems.	Carnegie Mellon University [USA]
Intelligent building control systems for thermal comfort and energy-efficiency: A systematic review of artificial intelligence-assisted techniques [10]	Systematic review of AI in building control	Valuable for understanding AI applications	Reviews various AI-assisted techniques for building control, offering valuable insights into innovative methods and their applications in achieving thermal comfort and energy efficiency.	Smart Systems Laboratory [Morocco]
Deep Comfort: Energy-Efficient Thermal Comfort Control in Buildings Via Reinforcement Learning [25]	Energy-efficient comfort control	Important for residential thermal comfort	Utilizes reinforcement learning to achieve energy-efficient thermal comfort control in buildings, presenting a novel approach to balancing comfort and energy use.	Nanjing University of Science and Technology [China]

Table 1. *Cont.*

Title	Contribution	Importance	Novelty	Institution
Towards optimal control of air handling units using deep reinforcement learning and recurrent neural network [26]	Optimal control of air handling units	High impact on HVAC systems	Introduces a novel method combining deep reinforcement learning and recurrent neural networks for optimizing air handling units, providing an advanced approach to building control.	New York University [USA]
OCTOPUS: Deep reinforcement learning for holistic smart building control [27]	Holistic approach to building control	Significant for smart building management	Employs deep reinforcement learning for comprehensive smart building control, offering an innovative approach to managing various aspects of building systems holistically.	University of California [USA]
Model input selection for building heating load prediction: A case study for an office building in Tianjin [28]	Model input selection for heating prediction	Important for accurate heating load forecasting	Proposes a new methodology for selecting model inputs in heating load prediction, providing a novel approach to improving prediction accuracy for building heating systems.	Tianjin University [China]
Comparison of machine learning models for occupancy prediction in residential buildings using connected thermostat data [29]	Comparison of ML models for occupancy	Important for improving occupancy predictions	Compares various machine learning models for predicting occupancy using thermostat data, offering valuable insights into the effectiveness of different algorithms.	University of Toronto [Canada]
Building HVAC scheduling using reinforcement learning via neural network based model approximation [30]	Scheduling HVAC systems using reinforcement learning	Crucial for efficient HVAC scheduling	Applies reinforcement learning with neural network model approximation for HVAC scheduling, presenting a novel approach to optimizing building energy management.	University of Southern California [USA]
Occupancy-based HVAC control systems in buildings: A state-of-the-art review [31]	Comprehensive review of occupancy-based control	Important for understanding current technologies	Reviews state-of-the-art occupancy-based HVAC control systems, summarizing recent advancements and providing a comprehensive overview of current technologies.	Concordia University [Canada]

Table 1. *Cont.*

Title	Contribution	Importance	Novelty	Institution
Deep-learning-based fault detection and diagnosis of air-handling units [32]	Fault detection in HVAC systems	Important for HVAC maintenance and efficiency	Utilizes deep learning techniques for fault detection and diagnosis in air-handling units, offering a novel approach to improving maintenance and system reliability.	National Taipei University of Technology [China]
A novel deep reinforcement learning based methodology for short-term HVAC system energy consumption prediction [33]	Short-term prediction of HVAC energy consumption	Useful for short-term energy management	Introduces a novel deep reinforcement learning methodology for predicting short-term HVAC energy consumption, providing a sophisticated approach to energy forecasting.	Zhengzhou University [China]
Practical implementation and evaluation of deep reinforcement learning control for a radiant heating system [34]	Practical implementation of DRL for heating	Important for radiant heating control	Evaluates the practical implementation of deep reinforcement learning for controlling radiant heating systems, offering a novel and practical approach to system optimization.	
Learning-based CO_2 concentration prediction: Application to indoor air quality control using demand-controlled ventilation [35]	CO_2 prediction for air quality control	Important for indoor air quality management	Uses learning-based methods for predicting CO_2 concentrations to enhance indoor air quality control, presenting a novel approach to demand-controlled ventilation.	Indiana University-Purdue University Indianapolis [USA]
Can HVAC really learn from users? A simulation-based study on the effectiveness of voting for comfort and energy use optimization [36]	Simulation study on user interaction with HVAC	Relevant for user-centric HVAC optimization	Simulates user voting for HVAC control and energy optimization, offering a novel approach to understanding user preferences and their impact on system performance.	Universidade de Lisboa [Portugal]
Application of deep Q-networks for model-free optimal control balancing between different HVAC systems [37]	Model-free HVAC control with deep Q-networks	Significant for balancing multiple HVAC systems	Applies deep Q-networks for model-free control balancing different HVAC systems, introducing a novel method for optimizing complex HVAC controls.	College of Engineering, Seoul National University [Republic of Korea]

Table 1. *Cont.*

Title	Contribution	Importance	Novelty	Institution
Improving Energy Consumption of a Commercial Building with IoT and Machine Learning [38]	IoT and ML for improving energy consumption	Important for integrating IoT in buildings	Combines IoT and machine learning to improve energy consumption in commercial buildings, providing a novel approach to energy management through advanced technologies.	COMSATS Institute of Information Technology [Pakistan]
A deep reinforcement learning approach to using whole building energy model for HVAC optimal control [39]	Deep reinforcement learning for HVAC control	High impact on building energy management	Applies deep reinforcement learning to a whole building energy model for HVAC control, offering an advanced and novel approach to optimizing building systems.	

In contrast, non-peer-reviewed articles were omitted from consideration. Following the removal of duplicates, a total of 401 articles remained. These articles then underwent a screening process, in which we explicitly evaluated their abstracts to determine their relevance to the research questions outlined in Section 2.1. Subsequently, the full text of 267 relevant articles was meticulously analyzed to assess their quality and eligibility. A thorough review led to the exclusion of 16 studies that failed to address the research questions adequately. Our selection criteria also accounted for excluding articles that focused solely on AI methods in energy forecasting, indoor comfort, occupancy detection, control strategies, and fault detection without addressing the enhancement of HVAC energy management. Furthermore, articles were excluded if they demonstrated poor writing quality, needed more methodological clarity, inadequately explained their approaches, or presented insignificant results.

3. Scientometric Analysis

3.1. Publication Source

Journal articles are widely regarded as the most indispensable form of scholarly publication in terms of sources of dissemination. Based on estimations, a total of 56 scientific journals presently exist. Consequently, Figure 3 exclusively showcases the six journals that have published the highest volume of scientific articles. Among these, Building and Environment, Energies, and Energy and Buildings are the top three journals, having produced 16, 16, and 13 articles, respectively. These journals can be considered valuable references in the domain of HVAC systems and the efficacy of building energy systems. Furthermore, the fact that these journals are edited by Elsevier and MDPI enhances their credibility and authority within their specialized fields.

3.2. Most Cited Publications

Furthermore, research was conducted to assess the exact quantity of citations each paper obtained. A selection of over twenty citations from a total of carefully chosen articles that included both targeted research on energy saving and broad HVAC literature were included in the analysis. Examining each publication's contribution, importance, and novelty in Table 1 sheds light on new developments in the area. It explains why some studies have attracted a lot of interest and citations. A significant finding of this analysis is

the close relationship between a research paper's citation count and uniqueness. Higher citation counts are typically seen in papers that present new uses of sophisticated machine learning methods, such as generative adversarial networks (GANs), deep reinforcement learning, and gradient boosting. Research that pushes the frontiers of what is feasible in the subject, offers new methodology, or dramatically improves current approaches is highly valued by the academic community based on this pattern.

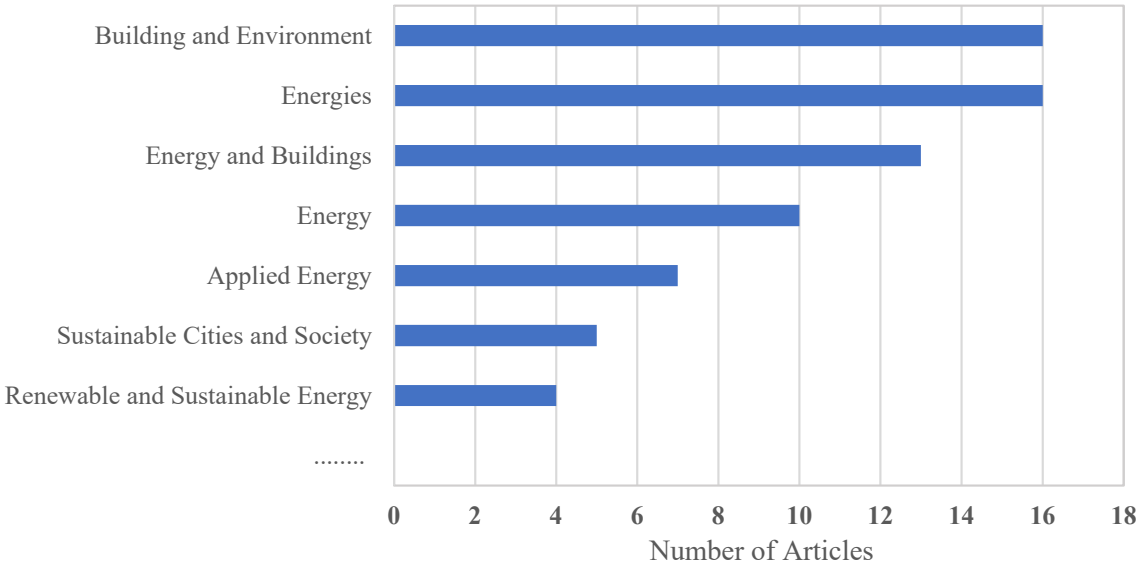


Figure 3. Publication sources.

It also emphasizes how crucial real-world applications are to maximizing the influence of research. Many of the most referenced publications deal with practical issues and provide answers that may be used immediately to build system management. Papers on occupancy-based control techniques, HVAC system defect detection, and predictive modeling for energy consumption, for instance, are essential because they address the pressing concerns of engineers and building managers. In addition to adding to the body of knowledge, these studies offer resources and techniques that may be applied to save expenses, increase energy efficiency, and improve the dependability of building systems. Additionally, the data demonstrate that more attention is often given to research that connects theoretical developments with real-world applications. Research and development efforts in energy management are frequently guided by studies that effectively integrate state-of-the-art AI and machine learning approaches. The studies that present novel frameworks for HVAC control—like those that use deep reinforcement learning—clarify this. They present fresh approaches to optimizing energy usage while preserving comfort and efficiency.

In conclusion, Table 1 shows how the uniqueness of the approaches presented and their applicability to real-world situations are directly related to the influence and significance of research in the field of energy management. Papers that present novel, creative solutions to practical issues are typically mentioned more frequently, which is a testament to their importance in advancing the discipline and benefiting society in concrete ways. This review emphasizes the importance of research that advances theoretical understanding while providing workable solutions for building energy system management.

3.3. Keywords Co-Occurrence

In order to investigate keywords with multiple interpretations, a thorough examination was conducted using varying search combinations, such as "Machine AND Learning", "Deep AND Learning", and "AI", along with "Buildings AND Energy AND HVAC".

Subsequently, a network analysis was implemented to establish the interconnectedness among 27 frequently appearing keywords out of the total pool of 832 identified keywords in VOSviewer 1.6.18. The resulting visual representation in Figure 4 depicts the frequency of keyword occurrences through node size, while the thickness of the links illustrates the degree of co-occurrence between the keywords. In the constructed network, comprised of three main entities, namely "Machine Learning", "HVAC", and "Energy Efficiency", the keyword co-occurrence network reveals notable associations between keywords like "Thermal Comfort", "Artificial Intelligence", "Deep Learning", and "Smart Buildings" with the keyword "Machine Learning". This correlation signifies that enhancing efficiency can be achieved by implementing machine learning techniques to assess its influence on the domains represented by these associated keywords. This analysis extends beyond theoretical advancements, offering practical implications for enhanced building codes and energy policies. Stakeholders, including policymakers and technology developers, are poised to benefit from these insights through improved regulatory frameworks and operational efficiencies.

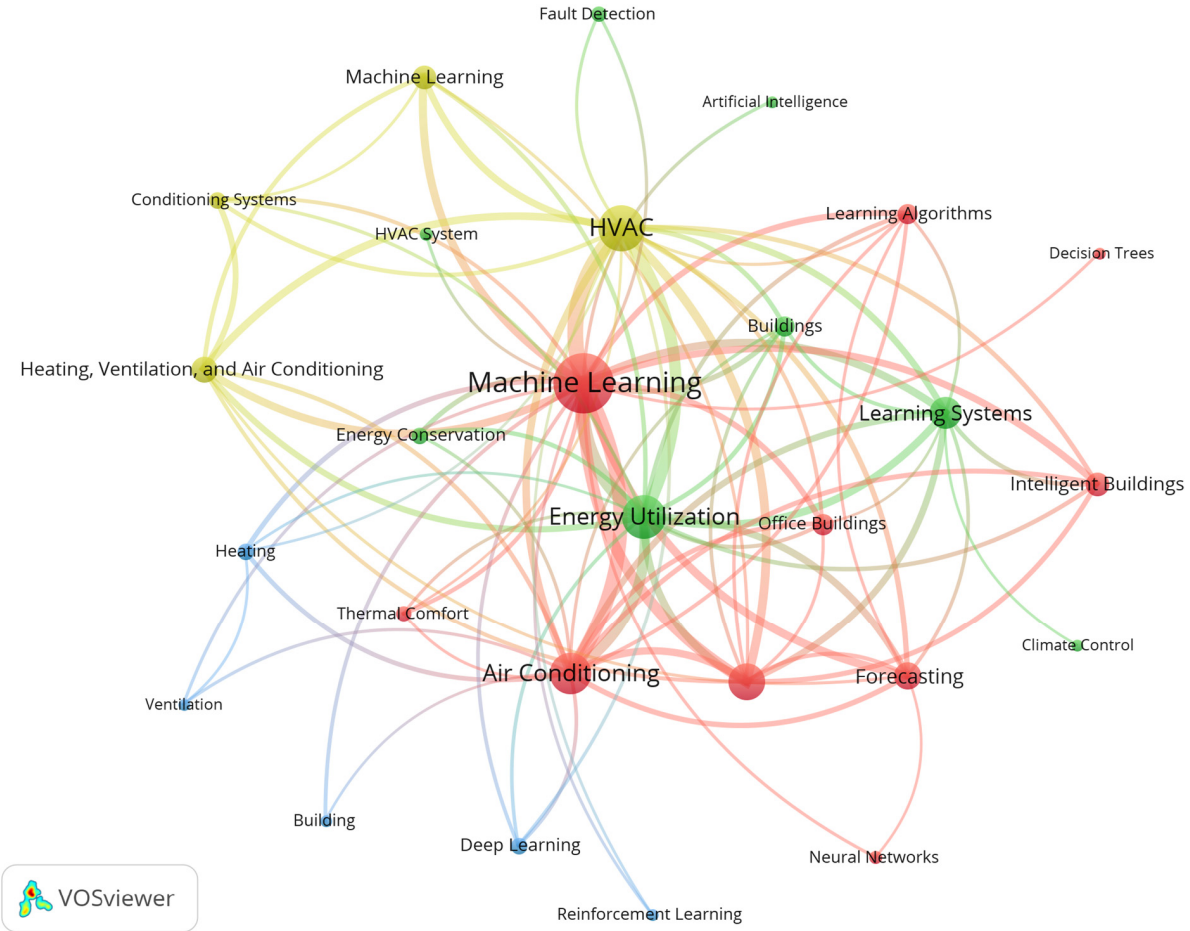


Figure 4. Keyword co-occurrence network.

3.4. Co-Authorship

The network visualization in Figure 5 displays co-authorship connections among countries, with various colors representing different connections. A major hub of this network is the USA, which indicates its extensive collaborative connections with numerous other nations. Lines between each country's nodes show co-authorship links in academic or research publications.

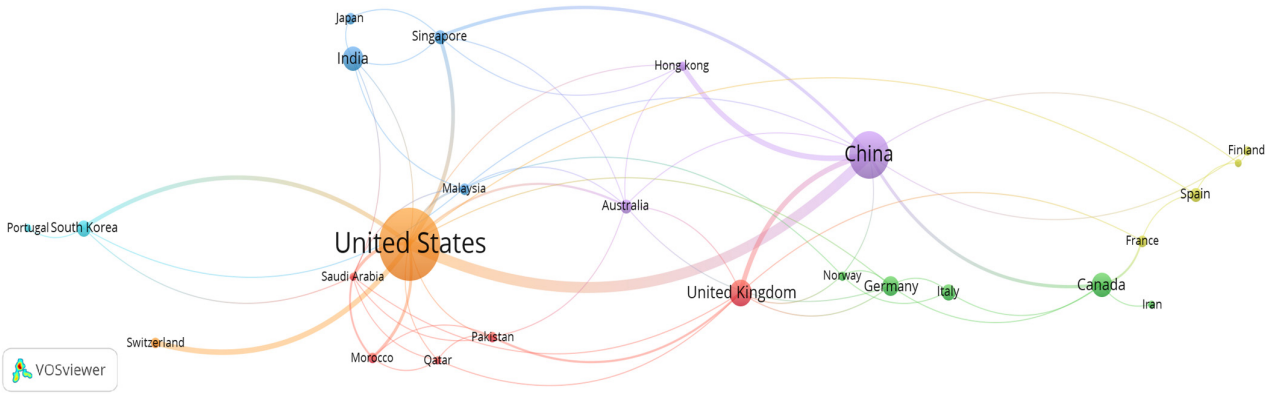


Figure 5. Co-authorship connections among countries.

There is a high volume of international collaborations between countries like the USA, China, and the UK. Different colors may be used to represent different regions or levels of collaboration. Among European countries, such as Germany, France, and the UK, connections could be displayed in one color. In contrast, connections between Asian countries, such as China, Japan, and Republic of Korea, could be displayed in another. There is likely a correlation between the thickness of the lines and the volume or frequency of collaborations, with thicker lines indicating stronger or more frequent collaborations. By identifying which countries are central in international research networks and the nature of their connections, the network effectively illustrates the global landscape of academic collaborations. The authors' varied backgrounds offer insightful local knowledge, guaranteeing that the study encompasses a broad spectrum of viewpoints on AI applications in HVAC systems. This study improves the generalizability of its findings across a range of climatic conditions and building types by integrating knowledge from diverse geographic locations.

3.5. AI-Based Method

Based on empirical research, various AI methods have been studied and are divided into two broad categories: the control and maintenance of HVAC systems. These methods aim to automate the necessary processes for HVAC system operation through the assistance of AI. By reducing manual steps in energy management, these methods have the potential to optimize energy consumption and mitigate energy waste. Specifically, an analysis of Figure 6 reveals that most articles (82.93%) examined the control method of HVAC systems, whereas a smaller percentage (17.07%) focused on their maintenance technique. This suggests that the control method has received more attention in the literature. Various AI techniques are employed to control and maintain HVAC systems. Control strategy, occupancy detection, consumption forecasting, and thermal comfort analysis are utilized in the control method. In contrast, fault detection and diagnosis (FDD) and maintenance management techniques are employed in the maintenance method. Figure 6 further illustrates that many articles (110) have explored control strategies for HVAC systems using AI, highlighting its prevalence as a control technique. Consumption forecasting, another commonly used control technique, is explored in 40 articles. Additionally, FDD, the principal maintenance method technique, is discussed in 37 articles.

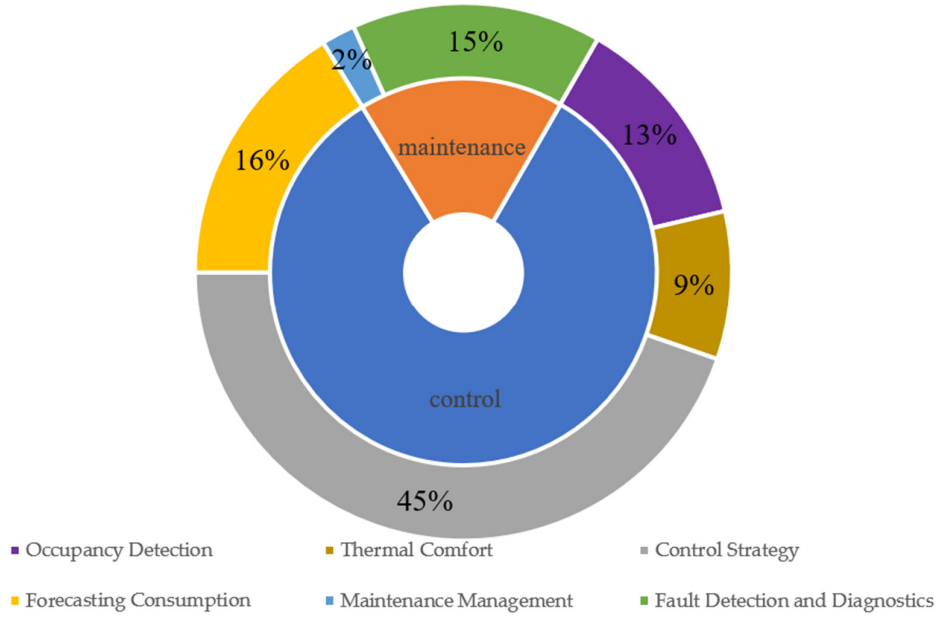


Figure 6. Number of articles per type of AI-based methods and technique.

This distribution of research focus indicates a clear trend favoring AI-driven control strategies over maintenance-related applications in HVAC systems. The dominance of control-focused studies suggests that researchers prioritize immediate energy efficiency improvements, adaptive control mechanisms, and real-time system optimizations, as these provide direct operational benefits. However, while fault detection and predictive maintenance remain less explored, they are crucial for long-term system reliability, cost reduction, and sustainability. The relatively lower number of studies on maintenance techniques highlights an opportunity for future research to develop more advanced AI-based maintenance frameworks, integrating real-time diagnostics, predictive analytics, and automated fault prevention to enhance the overall efficiency and longevity of HVAC systems.

3.6. AI Algorithms

Within the library, a combined number of 44 distinct algorithms are available. To depict the prevalence of these algorithms, Figure 7 presents the top 25 commonly utilized methods, categorized according to control and maintenance fields. Deep reinforcement learning (DRL) emerged as the most favored algorithm for control applications. Following suit, artificial neural network (ANN), Random Forest (RF), and long short-term memory (LSTM) algorithms ranked as the second, third, and fourth most commonly employed methods. Similarly, in maintenance approaches, Support Vector Machines (SVMs) and ANNs were identified as the predominant methods. Each algorithm has distinct advantages and disadvantages, making it suitable for specific tasks. A thorough examination of the primary benefits and drawbacks of several well-known algorithms used in HVAC control and maintenance applications can be found in Table 2. This detailed comparison serves as a valuable resource for selecting the most suitable algorithm for specific tasks across various domains. By understanding the strengths and limitations of each algorithm, practitioners can make informed decisions tailored to the unique requirements and constraints of their applications. This approach enables the effective and efficient implementation of AI models, enhancing the overall performance and management of HVAC systems.

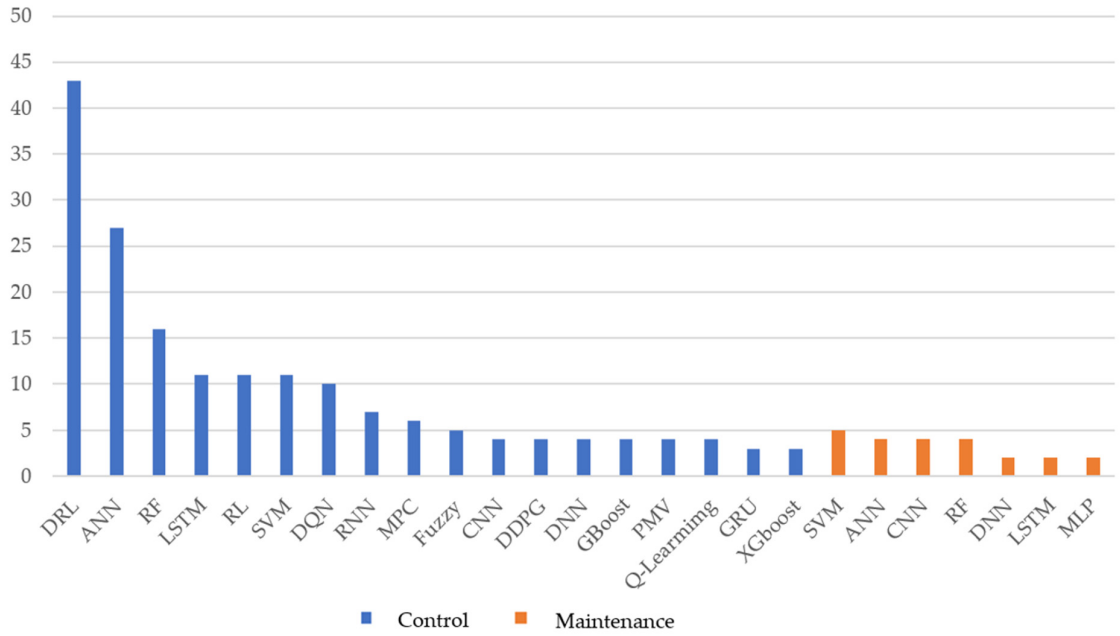


Figure 7. Number of articles per AI algorithm used.

Table 2. Advantages and disadvantages, associated techniques, and impacted components of each algorithm.

Algorithm	Advantages/Disadvantages	Technique	Component
ANN	Advantages: High accuracy in load forecasting; handles non-linear relationships. Disadvantages: Requires extensive training data; computationally intensive.	Forecasting Consumption Control Strategy Occupancy Detection Thermal Comfort	Chillers, Air Handlers [40] Compressors [41] Ventilation Systems [42] Duct Systems, Thermostats
SVM	Advantages: Effective for small datasets; robust to overfitting. Disadvantages: Not ideal for large datasets; kernel selection can be challenging.	Fault Detection and Diagnostics Occupancy Detection Maintenance Management	Airflow Systems [43] Air Quality Monitors [15] Compressors, Fans [44]
DRL	Advantages: Learns optimal control strategies without predefined models; adapts dynamically to uncertainties. Disadvantages: High complexity; slow convergence for large systems.	Control Strategy Forecasting Consumption Thermal Comfort	Entire HVAC System [45] Cooling Towers, Air Handlers [46] Thermostats, Sensors [47]
RF	Advantages: High accuracy for classification problems; resistant to overfitting. Disadvantages: May struggle with high-dimensional data; less interpretable.	Thermal Comfort Fault Detection and Diagnostics Forecasting Consumption	Thermostats [48] Ventilation Systems [49] Zone Equipment, Variable Air Volume (VAV) Systems [50]
LSTM	Advantages: Excellent for multivariate time series data; suitable for energy demand forecasting. Disadvantages: High memory requirements; slower training compared to traditional methods.	Forecasting Consumption Maintenance Management Thermal Comfort	Zone Temperature Controllers, Heat Pumps [40,51] Duct Systems, Heat Exchangers [44] Building Zones [52]

Figure 8, which presents a Sankey diagram, illustrates the flow and relationships between various AI methods, their techniques, and the specific algorithms used in HVAC systems for energy management. This visualization aids in understanding the predominant AI strategies that have been applied to control and maintenance operations within the

sector. It highlights the significant emphasis on control methods, particularly those utilizing deep reinforcement learning (DRL) and machine learning (ML) techniques, which are prominent in optimizing energy efficiency and operational effectiveness. The diagram also shows the linkage between these methods and key outcomes like energy consumption reduction and improved system reliability. By delineating these connections, Figure 8 provides crucial insights into how AI integration influences HVAC performance metrics and assists stakeholders in identifying areas of potential development or further research. This enhanced interpretation helps to bridge the gap between theoretical AI applications and practical HVAC system improvements, offering a clearer understanding of the strategic value of these technologies in building management.

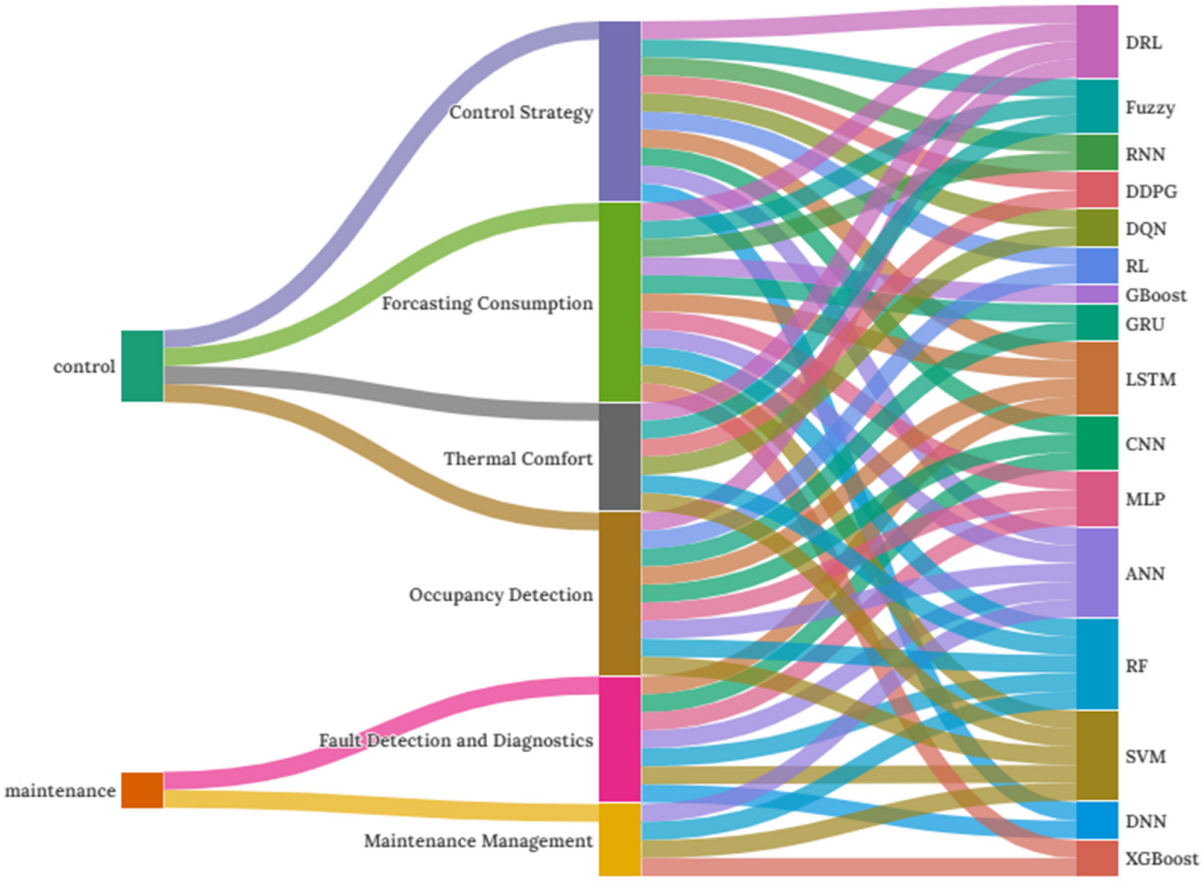


Figure 8. Sankey diagram between methods, techniques, and AI algorithms.

4. Systematic Literature Review

This section performs an exhaustive analysis utilizing the established library through a systematic literature review. After a comprehensive examination of these publications, significant subjects were clustered into primary informational classifications. These categories, their respective topics, and associated publications are exhibited in Table 3. AI models used for HVAC systems vary in complexity, precision, and computational needs. Model selection depends on data availability, real-time computing capacity, and system constraints [5,6]. Data quality issues, system integration, and fit to different building environments are also implementation challenges [7,8]. These challenges are key to delivering the successful implementation of AI-based HVAC systems [10,16].

Table 3. Categories and topics identified.

Categories	Topics	Publications
Control HVAC system	Control Strategy	[5,7,10,16,17,20–22,24,26,27,30,34,37–39,46,47,53–143]
	Occupancy Detection	[15,23,29,31,36,42,144–169]
	Forecasting Consumption	[14,18,28,33,45,50,51,170–202]
	Thermal Comfort	[25,35,41,49,203–220]
Maintenance HVAC system	Fault Detection and Diagnostics	[9,19,32,40,43,221–252]
	Maintenance Management	[12,44,253–255]

4.1. Control HVAC System

This category consolidates subjects about artificial intelligence methods employed in managing HVAC systems, which can potentially enhance energy efficiency within buildings. Within this category, the principal areas of focus include control strategy, occupancy detection, consumption forecasting, and thermal comfort assessment.

4.1.1. Control Strategy

The lack of intelligent and remote controls in managing chillers and air handlers within HVAC systems leads to significant energy consumption. However, the Internet of Things (IoT) has introduced technological advancements that facilitate remote control functionalities for these devices. Alternatively, the progress of machine learning (ML) and artificial intelligence (AI) allows devices to be trained to autonomously make informed decisions that influence interactions between humans and machines [56,57,106]. These advancements are supported by combining wireless sensor data or IoT with user feedback within an AI system. Model predictive control (MPC), which utilizes ML techniques to generate optimal schedules for HVAC systems, can be optimized and simulated using artificial neural networks (ANNs) [61,89,116]. Nonetheless, designing an optimal energy management algorithm for scheduling HVAC systems is challenging due to uncertainty in the models, parameters, and temporally linked operational constraints. The deep reinforcement learning (DRL) algorithm overcomes these challenges, as it does not require prior knowledge of uncertain parameters or the construction of a thermal dynamics model, making it suitable for energy management and improving user comfort [17,94,131,138]. A comparison between DRL and MPC is provided in a separate study [81]. The integration of IoT with AI presents promising advancements for HVAC systems. However, significant research gaps in scalability and user adaptability must be addressed to harness these technologies fully.

Modern HVAC systems rely on advanced control methods that incorporate machine learning (ML) techniques and fuzzy logic to achieve energy efficiency. For example, Elman neural networks have demonstrated significant potential in load forecasting by predicting HVAC system requirements based on dynamic environmental and occupancy data. These adaptive control strategies ensure systems respond promptly to changing conditions, reducing energy waste while maintaining performance levels. Additionally, hybrid models that combine ML and IoT functionalities allow more granular control of chillers and air handlers, ensuring improved coordination between system components [41,45].

AI-driven control strategies have demonstrated energy savings of up to 40% in various studies, depending on factors such as building type, climatic conditions, and AI model selection. For instance, reinforcement learning-based HVAC optimization has been shown to reduce energy consumption by 30–40% in smart buildings [16], while ANN-based predictive control strategies have achieved savings in the range of 20–35% [31].

4.1.2. Occupancy Detection

Building occupancy detection and estimation play a crucial role in enhancing building energy efficiency. Over the last ten years, numerous approaches have been suggested to enhance the precision of detecting and estimating occupancy [144,152,160,169]. Employing machine learning models and methodologies in predicting occupancy behavior and routines in buildings has spawned their application in the operational management of building systems [150,153]. In identified studies, machine learning techniques are employed to forecast patterns of occupancy and behavior, particularly emphasizing their potential implementation in building systems to enhance energy efficiency, indoor air quality, and thermal comfort. The model system and prediction timeframes are different, making it challenging to develop a perfect prediction model for building occupancy. In general, occupancy prediction models usually include data collection, occupancy prediction, and validation [145]. Despite the potentially higher prediction accuracy of ANNs, they may not possess the same level of transferability as traditional ML models [23,157,167]. In order to identify and classify real-time entities, a convolutional neural network (CNN) architecture was trained and implemented [146,151,165]. Long short-term memory (LSTM) can capture time series data. An attention model can accentuate their significance in energy prediction by allocating weights to the most pertinent inputs. Moreover, integrating the attention mechanism during the training of convolutional neural networks (CNNs) allows for decreased communication overhead. This mechanism effectively disregards extraneous information while amplifying pertinent data [154].

Advancements in AI have introduced innovative methods for precise occupancy detection in HVAC systems, such as using competitive learning and sensor fusion. These methods optimize the functionality of HVAC components like air handlers and ventilation systems by ensuring they align with actual occupancy needs. Furthermore, predictive models trained on historical occupancy and behavioral data can forecast usage patterns, enabling energy-efficient HVAC operation in both residential and commercial buildings [40,41].

4.1.3. Forecasting Consumption

Accurately estimating energy consumption in buildings, particularly within the HVAC system, which contributes significantly to electricity usage, is crucial for effective energy management [45,192]. In this regard, data-driven models have become extensively employed for precise energy consumption prediction [197].

Considering that HVAC accounts for a large portion of the electricity load in buildings, forecasting consumption is vital for energy management [42,187]. Data-driven models are widely used to predict energy consumption [197]. Knowing the properties of outdoor air and the number of individuals using a space enables the estimation of energy consumption of a cooling system in a building with a considerable level of precision [170]. In load forecasting, LSTM networks have demonstrated impressive outcomes [195]. These networks possess LSTM layers that explicitly grasp the temporal associations in forecasting multivariate time series, thus presenting significant potential in energy-efficient building technologies [198]. The recurrent neural network (RNN) model is employed to predict the thermal load of a building as well as the temperatures within its different zones [168,176,183,197].

Energy consumption forecasting is an essential aspect of modern HVAC systems, particularly for components like compressors and air conditioners, which contribute heavily to peak energy usage. Recurrent neural networks (RNNs) and predictive analytics platforms are increasingly utilized to forecast cooling and heating loads. These methods incorporate factors like weather patterns and occupancy schedules, enabling HVAC systems to preadjust their operations and optimize energy use in real time [40,45].

4.1.4. Thermal Comfort

Most individuals utilizing shared office spaces express discontent with their thermal comfort due to HVAC systems' inability to cater to individualized thermal environment needs [203,219]. By integrating considerations for both energy consumption and occupant thermal comfort, the regulation of thermal comfort can be approached as a problem of minimizing costs. One approach to predicting occupant thermal comfort involves the utilization of a Feedforward Neural Network (FNN) [217]. Additionally, certain studies employ an RF algorithm within HVAC systems to estimate the air quality index [202,204,217]. While AI significantly enhances HVAC efficiency, challenges such as system compatibility and data privacy remain prevalent. By using the environmental information, the HVAC system can be adjusted to predict thermal comfort in real time. Occupancy data, such as occupancy number and activity level, can be used to estimate indoor CO_2 and minimum ventilation levels. Thermal comfort prediction models also use occupancy and activity levels. This information can be used to optimize HVAC operations and minimize energy consumption [145].

Achieving optimal thermal comfort while minimizing energy consumption is a key challenge in HVAC design. Recent innovations involve using AI models, such as fuzzy logic controllers and reinforcement learning systems, to regulate key components like air ducts and chillers. These systems dynamically adjust HVAC parameters based on real-time environmental and user feedback, improving indoor air quality and occupant satisfaction [41,49].

4.2. Maintenance HVAC System

This classification aims to provide insights into the potential of AI and its algorithms in enhancing the energy efficiency of HVAC systems through maintenance practices. By exploring maintenance management concepts and FDD, it is feasible to reduce energy consumption while ensuring optimal comfort levels. This classification aims to illustrate some insights into the potential role of artificial intelligence and its algorithms in increasing the energy efficiency of HVAC systems by enhancing maintenance practices.

4.2.1. Fault Detection and Diagnostics

In recent years, scholars have intensified their efforts in advancing the study of automated fault detection and diagnostics, prompted by the rise in energy consumption attributable to building faults [228]. Such faults have led to suboptimal thermal comfort and reduced occupant productivity [231]. By investigating computing-based FDD methods, two distinct categories emerge: knowledge-based and data-driven approaches [18,19]. As a result of the insufficient number of fault samples, the imbalanced class classification problem, which is prominent in machine learning, makes it challenging to categorize faults [19,241]. Discrete Bayesian Networks (DisBNs) are employed to diagnose faults across different levels in commercial HVAC systems. The implementation of DisBNs addresses a common challenge faced by other FDD methods, namely the lack of sufficient fault data for training purposes. A DisBN considers the interactions between faults and fault symptoms (referred to as fault indicators) within HVAC systems, considering the impacts of faults at various levels [227]. To enhance the effectiveness of HVAC system detection and isolation, this study combines a multiscale representation, Principal Component Analysis (PCA), and ML classifiers [252]. By identifying faults and predicting future states, machine learning algorithms have the potential to aid in scheduling maintenance tasks [226].

Efficient fault detection and diagnostics are crucial for maintaining HVAC systems like compressors, chillers, and air handlers. AI-driven approaches, such as fuzzy inference systems and predictive maintenance frameworks, enable early identification of potential

faults. These methods leverage historical maintenance data and real-time monitoring to reduce downtime and operational costs, ensuring components remain in optimal condition. For example, predictive models applied to chiller plants can detect anomalies linked to external factors like weather fluctuations [48,49].

AI-driven fault detection and diagnosis (FDD) has significantly enhanced HVAC reliability by identifying system anomalies before they result in failures. Machine learning models, such as Random Forests and Bayesian Networks, are extensively used to detect irregularities in temperature, airflow, and energy consumption patterns [223].

Studies have demonstrated that predictive maintenance frameworks leveraging AI can reduce HVAC downtime and extend equipment lifespan by providing real-time fault detection alerts. These systems proactively diagnose potential failures, allowing facility managers to address issues before they escalate into costly repairs [9].

4.2.2. Maintenance Management

The role of the HVAC system in facility management and maintenance operations is of paramount importance, as any malfunction can result in substantial financial drawbacks. The implementation of DL techniques can enhance the overall performance of buildings by reducing energy consumption, facilitating maintenance planning, and overseeing equipment operations. Such improvements are particularly notable in the domain of predictive maintenance. Notably, significant advancements have emerged in building operation and maintenance, as evident in recent studies [12,44]. Upon analysis, extreme gradient boosting (XGB) has been identified as the most optimal machine learning algorithm for predictive maintenance [253].

Effective maintenance management ensures the longevity and efficiency of HVAC components, such as heat exchangers, evaporators, and ducts. AI-based predictive models, such as extreme gradient boosting (XGB), have been shown to excel in anticipating equipment failures and optimizing maintenance schedules. Furthermore, integrating digital twins with predictive analytics provides a real-time simulation of HVAC operations, allowing for proactive interventions to prevent component degradation [41,48].

AI-driven predictive maintenance strategies play a critical role in minimizing HVAC operational costs. By utilizing sensor-based monitoring and real-time analytics, AI models can detect wear and tear in HVAC components, triggering maintenance alerts before failures occur [255]. Cloud-based AI models have been implemented for large-scale HVAC maintenance optimization, allowing for remote diagnostics, automated scheduling of repairs, and performance tracking across multiple buildings. These intelligent maintenance strategies not only enhance system efficiency but also reduce energy waste and carbon footprints in smart buildings [254].

4.3. Machine Learning Model Applicability in HVAC Applications

Several machine learning (ML) models are applicable to some extent based on the specific HVAC application:

- Supervised Learning (e.g., decision trees, Support Vector Machines, neural networks): Best suited for fault detection, energy consumption forecasting, and HVAC system diagnostics [14,19].
- Unsupervised Learning (e.g., Clustering, Principal Component Analysis): Suitable for anomaly detection and system optimization with unlabeled data [23].
- Reinforcement learning (e.g., deep Q-Networks, Actor–Critic Models): Most appropriate for adaptive HVAC control, real-time dynamic setpoint changes, and real-time energy optimization [26,32].

The model selection depends on the computational complexity, data availability, and requirement for real-time processing [30].

5. Discussion

This study elucidates the potential for AI to revolutionize energy management in HVAC systems globally, offering substantial benefits across various sectors. Even though this review is not focused on individual case studies, it combines major findings from various AI applications in HVAC systems, providing a systematic overview of their impact, challenges, and future directions. For the HVAC industry worldwide, the application of machine learning and deep learning highlights significant opportunities for energy savings and operational efficiencies crucial for commercial buildings. These adaptations enhance system performance and adjustment to diverse climatic conditions, making them versatile solutions, applicable globally. Further, by demonstrating AI's effectiveness in reducing energy consumption and emissions, this research supports establishing stricter energy efficiency standards and regulations internationally. Such findings can guide policy decisions to encourage or mandate the integration of intelligent systems in new and existing structures, aiding in achieving energy conservation and emission reduction targets. From a financial perspective, leveraging AI in building energy systems presents a significant advantage by decreasing operational costs and boosting the sustainability profile of real estate assets. This shift can draw investments, particularly under the increasing focus on Environmental, Social, and Governance (ESG) criteria within investment practices.

In assessing the effectiveness of AI technologies in HVAC systems, it is crucial to acknowledge several inherent limitations that may influence the generalizability and applicability of our findings. Our analysis depends heavily on the data presented in previously published research. The variability and limited availability of detailed operational data can restrict the extent to which conclusions can be generalized across different real-world scenarios. This limitation is significant as data deficiencies may skew the perceived effectiveness of AI techniques in HVAC systems. Furthermore, the review extensively covers various AI methods applicable to HVAC energy management. However, the practical deployment of these techniques is often hindered by existing infrastructure and the technological readiness of specific regions or industries. This mismatch between theoretical potential and practical feasibility can significantly limit the immediate applicability of our findings. Additionally, the impact of AI-driven solutions on energy efficiency and system performance can vary markedly depending on building architecture and climatic conditions. The studies reviewed may need to adequately represent this diversity, potentially limiting the reliability of AI solutions across different building types. These factors are critical as they not only influence the outcomes of our study but also affect the scalability and practical implementation of AI-driven energy management strategies.

Figure 9 presents a comprehensive overview of the current landscape, challenges, and future prospects of integrating AI in HVAC systems. The current landscape emphasizes the dominant role of AI-driven control methods and maintenance techniques in optimizing system performance and energy efficiency. However, significant challenges, such as infrastructure compatibility, scalability issues, data privacy concerns, integration with legacy systems, and the lack of long-term effectiveness studies, create barriers to wider adoption. Despite these hurdles, we can envision a future landscape in which there is a transformative shift with advancements in IoT integration, adaptive algorithms, customization for regional climatic conditions, and zero-carbon energy management, highlighting AI's potential to redefine HVAC operations sustainably and effectively.

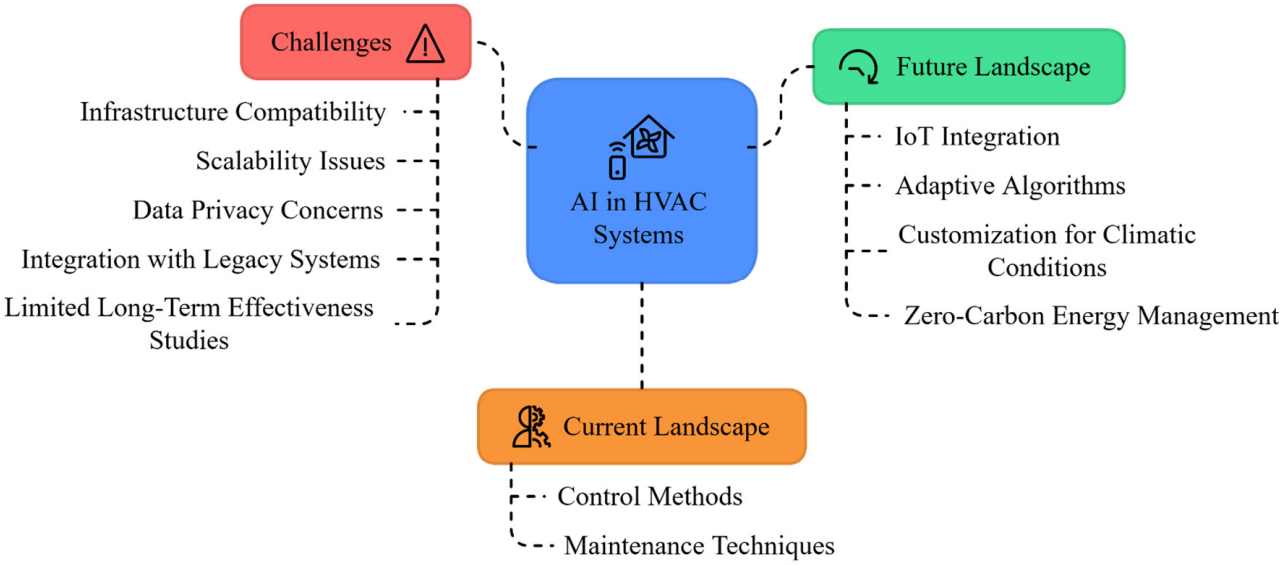


Figure 9. Overview of AI in HVAC Systems, highlighting the current landscape.

This review identifies several potential sources of error that could impact the validity and applicability of the findings. One such source is algorithmic bias, where AI models, heavily reliant on historical data, may not adequately capture the diversity of operational scenarios, building types, or climatic conditions. This can lead to biased predictions and management recommendations, affecting the accuracy of AI-driven solutions for HVAC systems. Another risk involves model overfitting, where complex AI algorithms such as deep learning models might overfit specific datasets used in studies. This tendency can hinder the algorithms' ability to generalize findings to other settings or adapt to evolving conditions due to climate change. Additionally, many studies utilize simulated data to train and validate AI algorithms. These simulations' assumptions can introduce errors, especially if they fail to accurately mimic real-world HVAC system behaviors and interactions with building occupants. These potential errors could undermine the effectiveness of the recommendations derived from this analysis, possibly leading to higher-than-expected energy usage or diminished occupant comfort. Such discrepancies have significant implications for policymakers and the scientific community, particularly in terms of meeting climate change targets. Erroneous predictions and suboptimal recommendations could impede the deployment of effective AI-driven energy management systems, thereby delaying progress toward crucial sustainability goals.

Even with the advancements made by AI-based HVAC systems, there are various gaps in the research area that need to be addressed. Apart from machine learning and deep learning approaches, scalability and implementation challenges hold back the widespread use of AI-based HVAC systems. Various AI models perform well in simulations but lack implementation capability due to computational complexity, integration problems, and varying building conditions. Furthermore, most existing AI solutions are designed for specific building classes or climate zones, making it difficult to develop generic models that can be implemented in various HVAC systems.

Another important challenge is the explainability and interpretability of AI models. Deep learning approaches, particularly neural networks, act as "black-box" models, reducing trust in AI-based HVAC control. Explainable AI (XAI) approaches need to be explored in future studies in order to improve model transparency and user acceptance.

Another limitation is real-time adaptation, as most AI-based HVAC models utilize historical data rather than learning in real time from varying environmental conditions.

Developing AI systems with real-time learning and adaptation would make them more efficient in uncertain environments.

In addition, integration with smart grids and renewable energy systems is yet to be fully investigated. AI systems must be able to coordinate HVAC control with energy demand-response planning and renewable energy supplies in order to optimize efficiency.

Finally, the lack of standardized, high-quality datasets for training AI models is a primary limitation. Most HVAC datasets contain missing, biased, or inconsistent information, limiting the reliability of AI-based predictions. In the future, more efforts should be dedicated to creating standardized, high-quality datasets for developing AI-driven HVAC control.

Future AI-driven HVAC systems should incorporate adaptive models for better energy efficiency. The selection of ML models should be guided by the specific application, balancing accuracy, computational cost, and real-time adaptability.

Study Significance

Rising energy needs and demands for efficiency and intelligent HVAC systems have motivated researchers to take up AI-powered systems. In the present work, a critical review of applications of AI in optimizing HVAC systems has been provided, with perspectives on energy efficiency, preventive maintenance, and adaptive control methods.

The usefulness of this study emanates from the following contributions:

- Closing the theory and practice gap of AI in HVAC by categorizing AI techniques and determining their suitability in real situations.
- Identifying the major challenges in AI-based HVAC implementation, including data availability, model interpretability, and integration with existing building management systems.
- Highlighting the strengths of machine learning (ML) and deep learning (DL) models in improving HVAC performance, reducing operating costs, and improving occupant comfort.
- Providing future research opportunities by emphasizing the significance of hybrid AI models, IoT-based smart HVAC systems, and more transparent AI-driven decision-making.

With the integration of the latest advances in AI for HVAC, this research is a valuable reference work for researchers and industry experts, guiding future developments in smart and sustainable building control.

6. Conclusions

This paper critically assesses the current literature on the use of artificial intelligence (AI) in the improvement of the energy efficiency of HVAC systems. The conclusions indicate that AI provides an opportunity to decrease energy usage and costs and, at the same time, improve the thermal comfort of the occupants. In particular, the integration of the machine learning and deep learning approaches has been identified as being capable of cutting energy consumption by as much as 40%, which presents a strong solution for energy management in buildings. The scientometric analysis reflects the increasing number of studies that concentrate on the application of AI to HVAC systems; it also indicates a focus on real-world usability. Nevertheless, our research also points to a major drawback in the form of the lack of research on long-term effectiveness and scenarios in various climatic conditions, which are recommended for future research.

This paper thus concludes that, while AI has the potential to revolutionize the HVAC industry, there are still issues that need to be addressed, such as standardization, data privacy, and integration with the current infrastructure. For policymakers and industry

leaders, these findings call for the development of legal and regulatory frameworks that will encourage the integration of AI technologies in the right manner and in a way that will facilitate the achievement of sustainable development goals and net-zero targets. In addition, the criticism in this article does more than amplify the importance of AI developments. Rather, it examines the real effects of these advancements and paves the way for future research that should concentrate on the implementation issues and costs of AI-based energy systems. By addressing these aspects, future research should aim to continue the exploration of AI capabilities and ensure these technologies are accessible, efficient, and effective across various regions and building types, thereby contributing to global efforts in energy sustainability.

The key findings are as follows:

- HVAC efficiency is improved significantly by the use of predictive control, supervision, and automated tuning thanks to the use of machine learning models.
- Some of the machine learning models skilled in fault diagnosis, predictive maintenance, and energy consumption shifting forecasting are artificial neural networks, decision trees, and reinforcement learning techniques.
- In comparison to other methods, deep reinforcement learning is more effective in the real-time controlling and regulating of HVAC systems with simultaneous adjustments in the environment and can lower the cost of operations.
- Even with the advanced technologies, data quality and the interpretability of the models in the context of integration of all system components is still a challenge to the dominance of AI in HVAC systems.
- Future research should focus on hybrid AI approaches, IoT integration, and explainable AI to enhance the practical application of AI-driven HVAC solutions.

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Abbreviations

The following abbreviations are used in this manuscript:

HVAC	Heating, Ventilation, and Air Conditioning
SLR	systematic literature review
BEM	building energy management
DRL	deep reinforcement learning
ANN	artificial neural network
RF	Random Forest
SVM	Support Vector Machine
IoT	Internet of Things
FDD	fault detection and diagnosis
MPC	model predictive control
CNN	convolutional neural network
LSTM	long short-term memory
RNN	recurrent neural network

GAN	generative adversarial network
WoS	Web of Science
XGB	extreme gradient boosting
PCA	Principal Component Analysis
DisBN	Discrete Bayesian Network
DQN	deep Q-Network
DDPG	Deep Deterministic Policy Gradient
XAI	Explainable Artificial Intelligence
GRU	Gated Recurrent Unit
MLP	Multi-Layer Perceptron

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